

MDS5202 Assignment 3

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Conceptual

5. We now examine the differences between LDA and QDA.
 - (a) If the Bayes decision boundary is linear, do we expect LDA or QDA to perform better on the training set? On the test set?
 - (b) If the Bayes decision boundary is non-linear, do we expect LDA or QDA to perform better on the training set? On the test set?
 - (c) In general, as the sample size n increases, do we expect the test prediction accuracy of QDA relative to LDA to improve, decline, or be unchanged? Why?
 - (d) **True or False:** Even if the Bayes decision boundary for a given problem is linear, we will probably achieve a superior test error rate using QDA rather than LDA because QDA is flexible enough to model a linear decision boundary. Justify your answer.
 6. Suppose we collect data for a group of students in a statistics class with variables X_1 = hours studied, X_2 = undergrad GPA, and Y = receive an A. We fit a logistic regression and produce estimated coefficients, $\hat{\beta}_0 = -6$, $\hat{\beta}_1 = 0.05$, $\hat{\beta}_2 = 1$.
 - (a) Estimate the probability that a student who studies for 40 h and has an undergrad GPA of 3.5 gets an A in the class.
 - (b) How many hours would the student in part (a) need to study to have a 50% chance of getting an A in the class?
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Applied

Data Loading Instructions:

To complete the following questions, please ensure you have the [ISLR](#) (or [ISLR2](#)) package installed. Question 13 uses the [Weekly](#) dataset.

```
# Load the required library and data
library(ISLR)
data(Weekly)
summary(Weekly)
?Weekly # Read the documentation
```

13. This question should be answered using the [Weekly](#) data set, which is part of the [ISLR](#) package. This data is similar in nature to the [Smarket](#) data from this chapter's lab, except that it contains 1,089 weekly returns for 21 years, from the beginning of 1990 to the end of 2010.
 - (a) Produce some numerical and graphical summaries of the [Weekly](#) data. Do there appear to be any patterns?

- (b) Use the full data set to perform a logistic regression with `Direction` as the response and the five lag variables plus `Volume` as predictors. Use the `summary()` function to print the results. Do any of the predictors appear to be statistically significant? If so, which ones?
- (c) Compute the confusion matrix and overall fraction of correct predictions. Explain what the confusion matrix is telling you about the types of mistakes made by logistic regression.
- (d) Now fit the logistic regression model using a training data period from 1990 to 2008, with `Lag2` as the only predictor. Compute the confusion matrix and the overall fraction of correct predictions for the held out data (that is, the data from 2009 and 2010).
- (e) Repeat (d) using LDA.
- (f) Repeat (d) using QDA.
- (g) Repeat (d) using KNN with $K = 1$.
- (h) Repeat (d) using naive Bayes.
- (i) Which of these methods appears to provide the best results on this data?
- (j) Experiment with different combinations of predictors, including possible transformations and interactions, for each of the methods. Report the variables, method, and associated confusion matrix that appears to provide the best results on the held out data. Note that you should also experiment with values for K in the KNN classifier.

15. This problem involves writing functions.

- (a) Write a function, `Power()`, that prints out the result of raising 2 to the 3rd power. In other words, your function should compute 2^3 and print out the results.
Hint: Recall that x^a raises x to the power a . Use the `print()` function to output the result.
- (b) Create a new function, `Power2()`, that allows you to pass any two numbers, x and a , and prints out the value of x^a . You can do this by beginning your function with the line:

```
> Power2 <- function(x, a) {
```

You should be able to call your function by entering, for instance,

```
> Power2(3, 8)
```

on the command line. This should output the value of 3^8 , namely, 6,561.

- (c) Using the `Power2()` function that you just wrote, compute 10^3 , 8^{17} , and 131^3 .
- (d) Now create a new function, `Power3()`, that actually returns the result x^a as an R object, rather than simply printing it to the screen. That is, if you store the value x^a in an object called `result` within your function, then you can simply `return()` this result, using the following line:

```
return(result)
```

The line above should be the last line in your function, before the `}` symbol.

- (e) Now using the `Power3()` function, create a plot of $f(x) = x^2$. The x-axis should display a range of integers from 1 to 10, and the y-axis should display x^2 . Label the axes appropriately, and use an appropriate title for the figure. Consider displaying either the x-axis, the y-axis, or both on the log-scale. You can do this by using `log = "x"`, `log = "y"`, or `log = "xy"` as arguments to the `plot()` function.

- (f) Create a function, `PlotPower()`, that allows you to create a plot of x against x^a for a fixed a and for a range of values of x . For instance, if you call

```
> PlotPower(1:10, 3)
```

then a plot should be created with an x-axis taking on values $1, 2, \dots, 10$, and a y-axis taking on values $1^3, 2^3, \dots, 10^3$.